

Exact Kernel Transfer to Clique Complexes: Hardness of Normalized Persistence and Gapped Betti Numbers

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Abstract

For an inclusion of clique complexes $X_1 \subseteq X_2$, the normalized persistence in degree d is the fraction $\text{rank}[H_d(X_1) \rightarrow H_d(X_2)] / \dim H_d(X_1)$ of the degree- d holes of X_1 that survive in X_2 . Lowe, Kim, Bondesan, and Hayakawa asked for the complexity of estimating this fraction under inverse-polynomial spectral-gap promises on both endpoint Laplacians, conjectured hardness for a variant SDQC_1 of DQC_1 with perfect completeness, and reduced the conjecture to an open kernel-preserving realization of history Hamiltonians by clique complexes. We carry out that realization for exact circuits over $G_2 = \{X, CX, CCX, H \otimes H\}$ and draw three consequences, all valid for unweighted clique complexes. (1) Normalized persistence is hard for the perfect-subspace-fraction class SF^{G_2} , which contains $\text{BQP}_1^{G_2}$ and the fraction form of $\text{SDQC}_1^{G_2}$ used in the hardness proofs of Lowe et al.; the hard instances have 2^r initial holes, of which the fraction $\dim S_x / 2^r$ survives, where S_x is the perfectly accepted subspace of the source verifier. (2) The hard instances satisfy the large-overlap and state-preparation conditions under which Lowe et al. place the problem in BQP , so on this family the problem is $\text{BQP}_1^{G_2}$ -hard and in BQP . (3) Computing the exact Betti number of a clique complex promised to have an inverse-polynomial spectral gap is $\#P$ -complete under weakly parsimonious reductions. The engine is a finite-certificate transfer theorem. For a fixed palette of weighted clique gadgets satisfying finitely many exactly checkable local conditions, every unit chain x of the full geometric complex satisfies

$$\text{dist}(x, K)^2 \leq C \left(t\lambda^2 + \frac{\langle x, \Delta x \rangle}{g\lambda^{2p}} \right),$$

where K is the embedded kernel of the simulated projector Hamiltonian, g its gap, t the number of gadgets, λ the private vertex weight, and p the private depth of the certified fillings ($p = 5$ here). Hence the geometric Laplacian has exactly $\dim K$ zero modes, a gap linear in g above them, and a harmonic space within $O(\lambda\sqrt{t})$ of K ; exact fillings identify the endpoint homology with a quotient V/W_A of the register cycle space, and nested term sets induce the natural quotient epimorphisms, so the persistent rank equals the later kernel dimension without any choice of compatible harmonic representatives.

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1 Introduction

Persistent homology records which topological features of a growing simplicial complex survive as the complex grows [8, 26, 5]. For a two-step filtration $X_1 \subseteq X_2$ and a degree d , the basic invariant is the persistent Betti number, the rank of the induced map $H_d(X_1) \rightarrow H_d(X_2)$. Dividing it by $\beta_d(X_1)$ gives a scale-free and directly interpretable statistic,

$$q_d(X_1, X_2) = \frac{\text{rank}[H_d(X_1) \rightarrow H_d(X_2)]}{\beta_d(X_1)} \in [0, 1], \quad (1)$$

the fraction of the degree- d holes of X_1 that are still present in X_2 . Lowe, Kim, Bondesan, and Hayakawa [20] recently introduced this quantity under the name *normalized harmonic persistence* and asked for its complexity on clique complexes of weighted graphs, the input model of quantum topological data analysis [19, 11, 9, 2, 22]. Their motivation is that the known quantum algorithms for Betti and persistent Betti numbers return quantities normalized by the number of simplices [19, 11, 25, 1], whereas a fraction of holes is the statistic one would actually ask for. Under inverse-polynomial spectral-gap promises on the two endpoint Laplacians they proved that a low-energy relaxation of q_d is DQC_1 -hard and in BQP , conjectured that q_d itself is hard for a perfect-completeness variant SDQC_1 of DQC_1 (their Conjecture 1), and reduced this conjecture to a *coefficient-sensitive kernel realization* of history Hamiltonians by clique complexes (their Conjecture 2). They describe the missing ingredient, a kernel-preserving simulation of frustration-free Hamiltonians by combinatorial Laplacians, as an important open question.

Why exact kernels are hard to realize.

The exact-kernel primitive available in the literature is the clique-gadget construction of Crichigno and Kohler [6] and King and Kohler [16]: a graph whose clique complex has one hole per computational basis state, together with one weighted gadget per local projector of a frustration-free Hamiltonian H . The gadgets produce the desired zero modes, and the many-gadget spectral analysis of King and Kohler proves a gap when the simulated H is positive definite. A persistence reduction needs more. It needs two Hamiltonians H_{in} and $H_{\text{out}} = H_{\text{in}} + H_{\text{rej}}$ with known but generally large kernels, the exact multiplicity $\dim \ker \Delta = \dim \ker H$ at both endpoints, an inverse-polynomial gap above the *complete* kernel at both endpoints, and control of the inclusion-induced map. One must rule out every additional geometric zero mode and every small positive eigenvalue, including for chains supported almost entirely on private gadget vertices, with constants that do not depend on the exponentially large padded register.

Results.

We carry out this realization for the history Hamiltonians of exact circuits over the gate set $G_2 = \{X, CX, CCX, H \otimes H\}$. Following Rudolph [24], $\text{BQP}_1^{G_2}$ is the class of promise problems decided with perfect completeness by uniform exact G_2 circuits; by

his exact simulation theorem it contains BQP_1^G for every finite gate set G with entries in a cyclotomic field $\mathbb{Q}(\zeta_{2^k})$, for instance Clifford+ T . Our source, however, is more general. An SF^{G_2} verifier (Definition 5.3) is an exact G_2 circuit on clean qubits and r unconstrained qubits with a subspace S_x of the unconstrained register that is accepted with certainty, every state orthogonal to S_x accepted with probability at most $1/3$, and a promise $\dim S_x/2^r \geq a$ or $\leq b$. This is the perfect-subspace analogue of DQC_1 with the decision imposed on the fraction of perfectly accepted states, the quantity that Lowe et al. identify as the power of SDQC_1 and from which their hardness proofs decide the instance; it contains $\text{BQP}_1^{G_2}$ through a three-bit label trick, and it contains $\text{SDQC}_1^{G_2}$ itself for all threshold pairs with $b < (3a - 1)/2$.

► **Theorem 1.1** (Main theorem; Theorem 5.4 and Corollaries 5.5 and 7.2). *For every $a - b \geq 1/\text{poly}$ there is a deterministic polynomial-time reduction that maps an instance x of an $\text{SF}^{G_2}(a, b)$ problem to nested unweighted clique complexes $X_{\text{in}} \subseteq X_{\text{out}}$ on a common vertex set, a degree d , and rationals $\gamma_{\text{in}}, \gamma_{\text{out}} \geq 1/\text{poly}(|x|)$ such that the endpoint Laplacians have no eigenvalue in $(0, \gamma_{\text{in}})$ and $(0, \gamma_{\text{out}})$ respectively,*

$$\beta_d(X_{\text{in}}) = 2^r, \quad \text{rank}[H_d(X_{\text{in}}) \rightarrow H_d(X_{\text{out}})] = \dim S_x, \quad q_d(X_{\text{in}}, X_{\text{out}}) = \frac{\dim S_x}{2^r}.$$

Consequently normalized harmonic persistence with inverse-polynomial endpoint gaps (Problem 9 of [20]) is SF^{G_2} -hard for weighted and for unweighted clique complexes; in particular it is $\text{BQP}_1^{G_2}$ -hard already when $\beta_d(X_{\text{in}}) = 8$ and $q_d \in \{3/4, 1/8\}$, and it is $\text{SDQC}_1^{G_2}$ -hard in the fraction form for all thresholds and in the trace form of [20, Definition 2] whenever $b < (3a - 1)/2$, with $\beta_d(X_{\text{in}}) = 2^{q-1}$ initial holes.

The second result concerns the algorithmic side. Lowe et al. show that the problem is in BQP when the uniform mixture over $\mathcal{H}_1 = \ker \Delta_{1,d}$ can be prepared and the pair $(P_{\mathcal{H}_2}, \mathcal{H}_1)$ satisfies a large-overlap condition (their Lemma 7). Both conditions hold on our instances.

► **Theorem 1.2** (Hard and in BQP ; Theorem 6.5 and Corollary 7.2). *The reduction of Theorem 1.1 can be arranged so that every nonzero eigenvalue of $P_{\mathcal{H}_{\text{out}}} P_{\mathcal{H}_{\text{in}}} P_{\mathcal{H}_{\text{out}}}$ is at least $1 - 1/\text{poly}$, and so that it outputs a polynomial-size circuit preparing the uniform mixture over $\mathcal{H}_{\text{in}}$ to trace distance $1/\text{poly}$. Normalized harmonic persistence therefore remains SF^{G_2} -hard when restricted to instances on which it is in BQP .*

The third result is a counting statement that uses only the classical part of the palette.

► **Theorem 1.3** (Gapped Betti numbers; Theorem 5.7 and Corollary 7.2). *Computing $\beta_d(\text{Cl}(G))$ for a graph G , weighted or unweighted, promised that $\Delta_{\text{Cl}(G),d}$ has no eigenvalue in $(0, \gamma)$ with $\gamma \geq 1/\text{poly}$, is $\#\text{P}$ -hard under parsimonious reductions and $\#\text{P}$ -complete under weakly parsimonious reductions; the same holds for persistent Betti numbers of gapped filtrations.*

Crichigno and Kohler prove $\#\text{P}$ -hardness of Betti numbers of clique complexes without a gap promise [6], Schmidhuber and Lloyd in the clique-dense regime, again

without a gap [25], and Cade and Crichigno with a gap for general chain complexes with local coboundaries [4]. Theorem 1.3 shows that the spectral gap which makes the simplex-normalized Betti number estimable by the quantum algorithms does not make the exact Betti number of a clique complex any easier.

The engine.

All three theorems rest on one transfer theorem and one quotient theorem.

► **Theorem 1.4** (Kernel transfer; Theorem 3.4, informal). *Fix a finite palette of weighted clique gadgets satisfying finitely many exactly checkable local conditions (Assumption 3.1), and let p be the largest number of private vertices in a simplex of the certified fillings. Attach t gadgets with private vertex weight λ to the register, simulating $H = \sum_j \Pi_j$ with $H \succeq g(P_V - P_K)$, $K = \ker H$. Then every unit chain x of the full clique complex satisfies*

$$\text{dist}(x, K)^2 \leq C \left[t\lambda^2 + \frac{\langle x, \Delta x \rangle}{g\lambda^{2p}} \right], \quad (2)$$

with C depending only on the palette. Hence, for $\lambda \leq c \min\{t^{-1}, \eta/\sqrt{t}\}$ and $E = c'\eta^2 g\lambda^{2p}$,

$$\dim \ker \Delta = \dim K, \quad \text{spec}(\Delta) \cap (0, E) = \emptyset, \quad \|P_{\ker \Delta} - P_K\| \leq \eta/2,$$

including the case $K = 0$.

Equation (2) concerns arbitrary geometric chains, not only vectors of the simulated register. Its shape matters. The structural error $t\lambda^2$ does not involve g , so the gadget scale λ is chosen from the number of gadgets alone, independently of the logical gap, and the resulting geometric gap E is linear in g ; a direct use of a many-gadget inequality in which λ is proportional to g would instead produce a high power of g . The only power of λ beyond λ^2 is λ^{2p} , which comes from a Schur-complement bound through the certified fillings and is independent of the locality of the gadgets; for our palette $p = 5$, and a conservative choice gives $E = \Omega(\eta^{12}g/t^{10})$. The projector bound is what yields Theorem 1.2.

► **Theorem 1.5** (Quotient naturality; Theorem 4.2). *Under the hypotheses of Theorem 1.4, $H_d(X_A) \cong V/W_A$ with $W_A = \sum_{j \in A} \text{ran } \Pi_j = K_A^\perp$, and for $A \subseteq B$ the inclusion $X_A \subseteq X_B$ induces the natural quotient map $V/W_A \twoheadrightarrow V/W_B$. In particular $\text{rank}[H_d(X_A) \rightarrow H_d(X_B)] = \dim K_B$.*

Theorems 1.4 and 1.5 give the two properties that Conjecture 2 of Lowe et al. requests, kernel fidelity with computable gaps and functoriality along the filtration, for the exact history Hamiltonians treated here; Section 8 states the precise comparison. Functoriality is obtained differently from the way it is formulated there. We never choose harmonic representatives compatibly across the filtration; exact fillings identify each homology group with a quotient of the fixed register cycle space, the induced map is then forced to be the natural quotient, and the projector bound shows a posteriori that the harmonic spaces are small rotations of the logical kernels.

Technical overview.

Five ideas carry the proofs.

Finite certificates and the zero-weight limit. In normalized coordinates the boundary pair of a gadget with private weight λ is affine in λ , $D(\lambda) = D_0 + \lambda D_1$, and D_0 is a fixed rational matrix. The certificate (Assumption 3.1) asks for four exact facts about each palette member: the boundary intersection $B_{d_a}(Y_a) \cap V_a$ equals the range of the intended rank-one projector (exact filling); $\ker D_0 = V_a \oplus Q_a$ for an explicit private subspace Q_a ; a projected private differential is coercive on Q_a at first order in λ ; and its retained outputs are private. All four are integer or modular linear algebra, and Appendix C records them for the palette of this paper.

Harmonic padding. A certificate speaks about a constant-size active register, while the actual gadget is joined with the unaffected register factors, so the padded complex is exponentially large. The join Laplacian is a tensor sum over bidegrees, and the unaffected factors have harmonics only in top degree. We keep the local private data tensored with these outside harmonics and place every nonharmonic outside mode in a sector with a constant gap (Lemma 3.2). This is the step where the constants become independent of the padded dimension.

Two energies and the filling depth. The proof of Equation (2) separates leakage out of the register from logical energy. Leakage is controlled by the zero-weight floor, the private coercivity, and an interface bound $\|[L_1 \cdots L_t]\|^2 = \|\sum_j L_j L_j^*\| \leq Ct\lambda^2$ that needs no orthogonality between the register outputs of different gadgets. Logical energy is controlled through $\Delta^\dagger \succeq c\lambda^{2p} H^{\text{emb}}$, which follows from the Cauchy–Schwarz inequality applied to the weighted certified fillings (Lemma A.1); the exponent is twice the number of private vertices in the deepest filling simplex.

From estimate to exact multiplicity and overlap. Every vector of the spectral subspace of Δ below E projects nontrivially onto K , so that subspace has dimension at most $\dim K$; exact fillings inject V/W_A into $H_d(X_A)$, providing $\dim K$ zero modes. Equality forces the exact multiplicity and the absence of eigenvalues in $(0, E)$. Applied at zero energy, the same estimate says that every harmonic vector is within $\eta/2$ of K ; equal dimensions then give $\|P_{\ker \Delta} - P_K\| \leq \eta/2$, and since $K_{\text{out}} \subseteq K_{\text{in}}$, the nonzero singular values of $P_{\mathcal{H}_{\text{out}}} P_{\mathcal{H}_{\text{in}}}$ are all at least $1 - \eta$.

Exact sources. Proposition 5.2 gives the two history kernels and gaps for any exact circuit with a perfectly accepted subspace and soundness $1/3$ on its complement, whatever the clean and unconstrained registers are. Three instances matter: a three-bit label turns any $\text{BQP}_1^{G_2}$ verifier into such a source with compressed acceptance operator $\text{diag}(1, p_x, p_x, p_x, p_x, p_x, 0, 0)$; a one-clean-qubit verifier is the SDQC_1 source; and a reversible evaluation of a Boolean formula gives the counting source, with S spanned by the satisfying assignments. Hayakawa’s common-copy blow-up [12], applied with the same labeled copy blocks at both levels, transfers everything to unweighted graphs.

Scope.

The hardness is gate-dependent, as in the known perfect-completeness hardness results for topological data analysis [6, 16, 10], and as in the definition of SDQC_1 itself. We

do not obtain ordinary BQP-hardness, gate-independent BQP₁-hardness, or SDQC₁-hardness in the trace form for threshold pairs with $b \geq (3a - 1)/2$; the last restriction is unavoidable for any reduction whose target sees only the fraction of perfectly accepted states (Appendix B.5). Complex phases and general integer states are outside the certified palette. The gap is a complexity promise rather than a practical estimate. Section 9 discusses these points and the open problems they suggest.

Organization.

Section 2 fixes conventions and the problem. Section 3 states the certificate model and the transfer theorem, Section 4 derives quotient naturality, Section 5 instantiates the history construction with a certified palette and proves the hardness and counting theorems, Section 6 proves the containment conditions, and Section 7 transfers everything to unweighted graphs. Section 8 compares with prior work and Section 9 concludes. The appendices contain the complete transfer proof, the history-gap and source calculations, and the finite palette certificates.

2 Preliminaries and problem statement

Weighted clique complexes.

All chain spaces are finite-dimensional over $\mathbb{C}$. A vertex-weighted simplicial complex has positive vertex weights $w(v)$, and a simplex has weight $w(\sigma) = \prod_{v \in \sigma} w(v)$. We use the normalized coordinates of King and Kohler [16], in which the weighted boundary of an oriented simplex removes a vertex v with coefficient $\pm w(v)$; in matrix form $\partial_{w,k} = W_{k-1}^{-1} \partial_{1,k} W_k$, where W_k is the diagonal matrix of k -simplex weights. The weighted complex is therefore conjugate to the unweighted one and has the same homology. We use

$$\Delta_{X,d} = \partial_{X,d}^* \partial_{X,d} + \partial_{X,d+1} \partial_{X,d+1}^*, \quad Z_d(X) = \ker \partial_{X,d}, \quad B_d(X) = \text{ran } \partial_{X,d+1}.$$

Finite-dimensional Hodge theory [7, 13] gives the orthogonal decomposition $C_d(X) = \mathcal{H}_d(X) \oplus B_d(X) \oplus \text{ran } \partial_{X,d}^*$ with $\mathcal{H}_d(X) = \ker \Delta_{X,d}$ and $Z_d(X) = \mathcal{H}_d(X) \oplus B_d(X)$, and identifies $H_d(X) = Z_d(X)/B_d(X)$ with the harmonic space $\mathcal{H}_d(X)$. For a positive semidefinite operator A , write $\gamma_+(A) = \min(\text{spec}(A) \setminus \{0\})$ for its smallest positive eigenvalue, with $\gamma_+(0) = +\infty$. All certified boundary matrices and fillings below are rational. Rank, kernel dimension, image intersection, and filling validity are unchanged under scalar extension $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$, and the real positive-semidefinite inequalities complexify verbatim. We state homology over $\mathbb{C}$ while certifying the finite palette over $\mathbb{Q}$.

Normalized persistence.

For $X_1 \subseteq X_2$, define

$$\beta_d(X_1 \rightarrow X_2) = \text{rank}[H_d(X_1) \rightarrow H_d(X_2)], \quad q_d(X_1, X_2) = \frac{\beta_d(X_1 \rightarrow X_2)}{\beta_d(X_1)}. \quad (3)$$

234 Lowe et al. [20] write the numerator through harmonic projectors, as the rank of
 235 $P_{\mathcal{H}_d(X_2)}$ restricted to $\mathcal{H}_d(X_1)$. The two formulations agree. For a cycle $z \in Z_d(X_2)$, the
 236 vector $P_{\mathcal{H}_d(X_2)}z$ is the harmonic representative of $[z]$, because $z - P_{\mathcal{H}_d(X_2)}z \in B_d(X_2)$;
 237 hence $z \mapsto P_{\mathcal{H}_d(X_2)}z$ induces the isomorphism $H_d(X_2) \cong \mathcal{H}_d(X_2)$. Since $\mathcal{H}_d(X_1) \subseteq$
 238 $Z_d(X_1) \subseteq Z_d(X_2)$ and $\mathcal{H}_d(X_1) \cong H_d(X_1)$, the operator $P_{\mathcal{H}_d(X_2)}|_{\mathcal{H}_d(X_1)}$ is the induced
 239 map $H_d(X_1) \rightarrow H_d(X_2)$ up to these identifications, and its rank is $\beta_d(X_1 \rightarrow X_2)$.

240 ► **Definition 2.1** (Gapped normalized persistence). *An instance of the weighted problem*
 241 *consists of explicit simple graphs $G_1 \subseteq G_2$ on one labeled vertex set V , one common*
 242 *positive dyadic weight function $w : V \rightarrow \mathbb{Q}_{>0}$, a degree $d \geq 1$, positive rational*
 243 *lower bounds γ_1, γ_2 , and a positive rational accuracy ε , all encoded in binary. Put*
 244 *$X_i = \text{Cl}(G_i)$, using the weighted-boundary convention above. The promises are*

$$245 \quad \beta_d(X_1) > 0, \quad \text{spec}(\Delta_{X_i,d}) \subseteq \{0\} \cup [\gamma_i, \infty) \quad (i = 1, 2).$$

246 *The task is to output $\tilde{q}$ such that $\Pr[|\tilde{q} - q_d(X_1, X_2)| \leq \varepsilon] \geq 2/3$. Input size is the*
 247 *total bit length of the graph encodings, weights, d , the two gap bounds, and ε . The*
 248 *unweighted problem is the restriction $w \equiv 1$. Its fixed decision version distinguishes*
 249 *$q_d \geq 3/4$ from $q_d \leq 1/8$.*

250 This is Problem 9 of Lowe et al. [20], with the same promises and with the encoding
 251 conventions fixed. The denominator is the initial Betti number, not the number of
 252 simplices, and the statistic counts exact zero modes rather than eigenvalues below a
 253 cutoff.

254 Register, logical projectors, and gadgets.

255 The construction starts from a top-dimensional *register* complex R , so $C_{d+1}(R) = 0$,
 256 and sets $V = Z_d(R) = \mathcal{H}_d(R)$. Concretely, R is the clique complex of the join of
 257 n bowtie graphs, one per logical qubit. A bowtie is the wedge of two 4-cycles; its
 258 clique complex is one-dimensional and connected with $\beta_1 = 2$, and the two normalized
 259 oriented petal cycles are the computational basis $|0\rangle, |1\rangle$. By the Künneth formula for
 260 joins, R has reduced homology only in degree $d = 2n - 1$, and $V \cong (\mathbb{C}^2)^{\otimes n}$ isometrically.
 261 Each constraint j is a rank-one projector π_j on the cycle space of a constant-size *active*
 262 sub-register; after tensoring with the identity of the unaffected factors it becomes an
 263 orthogonal projector Π_j on V , generally of higher rank. For a term set A , put

$$264 \quad H_A = \sum_{j \in A} \Pi_j, \quad K_A = \ker H_A, \quad W_A = \sum_{j \in A} \text{ran } \Pi_j = K_A^\perp, \quad (4)$$

265 and use a certified logical gap

$$266 \quad H_A \succeq g_A(P_V - P_{K_A}), \quad 0 < g_A \leq 1. \quad (5)$$

267 Each constraint is realized by a weighted clique gadget: a graph containing the
 268 active register, whose additional *private* vertices have weight λ and are joined to
 269 every vertex of the unaffected register factors, while all register vertices have weight
 270 one. Attaching the gadgets of all $j \in A$ to R gives the clique complex X_A , and Δ_A
 271 denotes its full Laplacian on all degree- d chains. Thus H_A acts on V , while Δ_A acts
 272 on $C_d(X_A) \supseteq V$.

► **Definition 2.2** (Independent private interiors). *The gadgets of a term set A have independent private interiors if private vertices of distinct gadgets are never adjacent. Every simplex of X_A that contains a private vertex then contains private vertices of exactly one gadget, and $C_d(X_A) = C_d(R) \oplus \bigoplus_{j \in A} C_{d,j}^{\text{priv}}$ as coordinate spaces.*

All constructions in this paper satisfy Definition 2.2: distinct gadgets receive disjoint sets of new vertices, and no edge is added between private vertices of distinct gadgets.

Circuit source.

For the circuit application, J inserts the fixed clean string around a maximally mixed logical input, U is an exact circuit, and

$$M = J^* U^* P_{\text{acc}} U J, \quad S = \ker(I - M).$$

The initial history kernel will have the full input dimension, while the final history kernel will be isomorphic to the perfect accepting space S .

3 Finite-certificate transfer to the complete geometric kernel

We isolate finite local conditions under which a clique-gadget construction controls every chain of the full geometric complex, including chains supported mostly outside the simulated register. The statement does not use convergence of a spectral projector whose rank grows with the simulated circuit.

3.1 The certificate model

Consider one active gadget before padding. Its weight-one register is R_a , its target cycle space is $V_a = Z_{d_a}(R_a)$, and every private vertex has weight λ . In normalized chain coordinates put

$$D_a(\lambda) = \begin{bmatrix} \partial_{d_a}(\lambda) \\ \partial_{d_a+1}(\lambda)^* \end{bmatrix} = D_{a,0} + \lambda D_{a,1},$$

so that $\Delta_{Y_a, d_a}(\lambda) = D_a(\lambda)^* D_a(\lambda)$. The decomposition is affine in λ because every entry of the normalized boundary is 0, ± 1 (a register vertex removed) or $\pm \lambda$ (a private vertex removed). The value at $\lambda = 0$ is a matrix limit, not a singular change of basis; $D_{a,0}$ is the boundary pair of a rational chain complex, since $\partial(\lambda)^2 = 0$ for all λ forces $\partial_0^2 = 0$.

► **Assumption 3.1** (Finite local certificate). *For every member a of a fixed finite palette there are an orthogonal subspace Q_a contained in the span of private simplex coordinates, an output projector R_a^{out} , and a rank-one projector π_a on V_a such that:*

- (i) $B_{d_a}(Y_a) \cap V_a = \text{ran } \pi_a$ for every $\lambda > 0$;
- (ii) $\ker D_{a,0} = V_a \oplus Q_a$, orthogonally;
- (iii) for $T_a(\lambda) = R_a^{\text{out}} D_a(\lambda)$,

$$T_a(\lambda) = \lambda T_{a,1}, \quad T_{a,1} V_a = 0, \quad \|T_{a,1} q\| \geq b \|q\| \quad (q \in Q_a);$$

307 **(iv)** every retained output coordinate contains a private vertex.
 308 The constant $b > 0$, the positive singular-value floor of $D_{a,0}$, and all matrix norms are
 309 uniform over the palette.

310 All data in Assumption 3.1 are finite integer or rational linear algebra. Exact fillings
 311 prove one inclusion in (i); matching boundary ranks and surviving register cosets,
 312 or the guard connecting-map argument, prove the reverse inclusion. Condition (i)
 313 does not depend on λ : the weighted boundary space is $W_{d_a}^{-1}B_{d_a}(Y_a; 1)$ and W_{d_a} is the
 314 identity on register coordinates. Modular rank bounds with matching rational kernels
 315 certify (ii), and a nonzero Gram determinant certifies (iii). The subspace Q_a need not
 316 be spanned by simplex coordinates; the selected-cycle guards of Appendix C produce
 317 Q_a of the form $Q \otimes \mathbb{C}\gamma$. The explicit palette used later is recorded in Appendix C.

318 ► **Lemma 3.2** (Harmonic padding). *Let $0 < \lambda \leq 1$. Let Y_a be joined with the unaffected*
 319 *register factors, whose join R_{out} has reduced homology only in its top degree, with top*
 320 *harmonic space $\mathcal{H}_{\text{out}}$, and let d be the global target degree. Then the zero-weight kernel*
 321 *of the padded gadget in degree d is*

$$322 \quad V \oplus (Q_a \otimes \mathcal{H}_{\text{out}}),$$

323 *its orthogonal complement has a uniform positive zero-weight floor, the projected pair*
 324 *$T_a(\lambda) \otimes P_{\mathcal{H}_{\text{out}}}$ satisfies (iii) with the same constant b , and the padded exact boundary*
 325 *intersection with V is the range of $\Pi_a = \pi_a \otimes I_{\mathcal{H}_{\text{out}}}$.*

326 **Proof.** See Appendix A.1. ◀

327 The harmonic restriction in this lemma is essential. If the padded bulk contained
 328 every coordinate with a local private vertex, deleting a weight-one outside vertex could
 329 remain in that bulk with coefficient one. We instead keep local private coordinates
 330 tensored with outside harmonics and place every nonharmonic outside factor in the
 331 separately gapped sector.

332 ► **Definition 3.3** (Filling depth). *Let $c_a \in C_{d_a+1}(Y_a; \mathbb{Z})$ be the certified integer filling of*
 333 *the integer register cycle v_a spanning $\text{ran } \pi_a$, so that $\partial c_a = v_a$ in unweighted coordinates.*
 334 *Its depth p_a is the largest number of private vertices in a simplex of the support of c_a ,*
 335 *and the depth of the palette is $p = \max_a p_a \geq 1$.*

336 The depth is a combinatorial invariant of the certificate and is read off from the filling
 337 data; Appendix C lists it for every member. It controls the only power of λ in the
 338 theorem: in weighted coordinates the filling becomes $W_{d_a+1}^{-1}c_a$, whose norm is at most
 339 $\|c_a\| \lambda^{-p_a}$.

340 3.2 Global theorem

341 Attach t_A padded gadgets to the common top-dimensional register R with independent
 342 private interiors (Definition 2.2). Let L_j be the block of $\partial_{X_A, d}$ that maps the private
 343 d -chains of gadget j to the register $(d-1)$ -chains. A private d -simplex has a register
 344 face only if it contains exactly one private vertex, and removing that vertex gives

coefficient $\pm\lambda$; hence every column of $\lambda^{-1}L_j$ has at most one nonzero entry and every row has at most p_j nonzero entries, where p_j is the number of private vertices of gadget j . Consequently the finite interface bound

$$L_j L_j^* \preceq N_j \lambda^2 I, \quad N_j \leq N_*, \quad (6)$$

holds with $N_j \leq p_j$, so N_* is a palette constant. The input blocks are orthogonal, while the outputs may overlap. Hence, for $L = [L_1 \ \cdots \ L_{t_A}]$,

$$LL^* = \sum_j L_j L_j^* \preceq N_* t_A \lambda^2 I.$$

► **Theorem 3.4** (Finite-certificate degenerate-kernel transfer). *Fix a finite palette satisfying Assumption 3.1 with depth p , and let $\kappa = 2p$. Let $\mathcal{A}$ be a finite family of term sets whose gadgets are padded palette members attached to the register R of Section 2 with independent private interiors, all with the same private weight λ . Let $t_{\max} = \max\{1, \max_{A \in \mathcal{A}} |A|\}$ and suppose*

$$H_A = \sum_{j \in A} \Pi_j \succeq g_A (P_V - P_{K_A}), \quad K_A = \ker H_A, \quad 0 < g_A \leq 1$$

for every $A \in \mathcal{A}$. There are palette constants $c_0, C > 0$ such that, whenever $0 < \lambda \leq \min\{1, c_0/t_{\max}\}$, every unit $x \in C_d(X_A)$ satisfies

$$\|(I - P_{K_A})x\|^2 \leq C \left[t_A \lambda^2 + \frac{\langle x, \Delta_A x \rangle}{g_A \lambda^\kappa} \right]. \quad (7)$$

Here P_{K_A} is the projector onto the embedded logical kernel, extended by zero outside the register.

For $0 < \eta < 1$, choose additionally

$$\lambda \leq c_1 \min\{t_{\max}^{-1}, \eta/\sqrt{t_{\max}}\}, \quad E_A \leq c_2 \eta^2 g_A \lambda^\kappa, \quad (8)$$

with palette constants $c_1, c_2 > 0$. Then

$$\dim \ker \Delta_A = \dim K_A, \quad \text{spec}(\Delta_A) \cap (0, E_A) = \emptyset. \quad (9)$$

This includes $K_A = 0$, in which case $\Delta_A \succeq E_A I$.

Proof architecture.

The full calculation is in Appendix A. The shared-output identity above and the private projected pairs first give the geometric leakage estimate

$$\|(I - P_V)x\|^2 \leq C \left[t_A \lambda^2 + (t_A + \lambda^{-2}) \langle x, \Delta_A x \rangle \right]. \quad (10)$$

The certified fillings give, by a Schur-complement argument (Lemma A.1),

$$\Delta_A^\uparrow \succeq c \lambda^{2p} H_A^{\text{emb}}, \quad (11)$$

where $\Delta_A^\uparrow = \partial_{d+1} \partial_{d+1}^*$ and H_A^{emb} is H_A extended by zero outside V . This is the only step in which a power of λ beyond λ^2 appears, which is why the exponent

depends on the fillings alone and not on the locality of the gadgets. Combining these inequalities with the logical gap proves Equation (7). The bound applies to every vector in a low-energy spectral subspace. Projection onto K_A is therefore injective on that subspace, so its dimension is at most $\dim K_A$. Independent exact fillings supply $\dim K_A$ zero modes through V/W_A ; equality and the whole-positive-spectrum gap follow. No relative-acyclicity assumption is used.

For the explicit history palette $p = 5$ (Appendix C), so $\kappa = 10$. Taking a dyadic $\lambda = \Theta(\eta/t_{\max})$ gives the conservative scale

$$E_A = \Omega\left(\frac{\eta^{12} g_A}{t_{\max}^{10}}\right). \quad (12)$$

For the certified atoms the bound $\Delta^\dagger \succeq c\lambda^{2p_a}\Pi$ is tight up to the constant (the optimal constant scales exactly as λ^{2p_a} numerically), so a better exponent would need gadgets with shallower fillings.

4 Filtered quotient naturality

The transfer theorem determines the dimension of every endpoint kernel. Exact filling determines the maps between those kernels at the homology level. This separation is useful: spectral concentration proves that there are no extra classes, while the quotient model computes persistence without comparing harmonic representatives.

► **Lemma 4.1** (Independent filling). *For every term set A ,*

$$B_d(X_A) \cap V = W_A = \sum_{j \in A} \text{ran } \Pi_j. \quad (13)$$

Proof. Every vector in $\text{ran } \Pi_j$ has a filling in gadget j , so $W_A \subseteq B_d(X_A) \cap V$. Conversely, write a degree- $(d+1)$ chain as a sum of its gadget components. If its total boundary lies in V , then its boundary has zero coordinate on every private d -simplex. Private supports of distinct gadgets are disjoint, so this vanishing holds separately for each gadget component. The remaining register boundary of component j lies in $B_d(Y_j) \cap V = \text{ran } \Pi_j$. Summing over j proves the reverse inclusion. ◀

No relative-acyclicity hypothesis appears in Lemma 4.1. Extra geometric homology outside V is excluded later by the all-chain concentration theorem, not by the boundary decomposition alone.

► **Theorem 4.2** (Natural quotient realization). *Under the hypotheses and parameter choice of Theorem 3.4, the map*

$$q_A : V/W_A \longrightarrow H_d(X_A), \quad q_A(v + W_A) = [v], \quad (14)$$

is an isomorphism. For $A \subseteq B$, these isomorphisms make the diagram

$$\begin{array}{ccc} V/W_A & \xrightarrow{\rho_{AB}} & V/W_B \\ q_A \downarrow \cong & & \downarrow \cong q_B \\ H_d(X_A) & \xrightarrow{(\iota_{AB})_*} & H_d(X_B) \end{array}$$

commute, where ρ_{AB} is the natural quotient map. Consequently,

$$\text{rank}(\iota_{AB})_* = \dim(V/W_B) = \dim K_B. \quad (15)$$

Proof. See Appendix A.6. ◀

For a chain $A_0 \subseteq A_1 \subseteq \dots \subseteq A_s$, every arrow in degree d is an epimorphism. The constructed barcode therefore has deaths but no new births in this degree. More generally, for any $i \leq j$,

$$\text{rank}[H_d(X_{A_i}) \rightarrow H_d(X_{A_j})] = \dim K_{A_j}.$$

Harmonic representatives may rotate between filtration levels; the exact statement concerns the algebraic homology maps.

Taking A_{in} to be the clock, propagation, and clean-input terms of a history Hamiltonian, and taking A_{out} to add the final rejection term, gives

$$q_d(X_{\text{in}}, X_{\text{out}}) = \frac{\dim K_{\text{out}}}{\dim K_{\text{in}}}. \quad (16)$$

This identity is the bridge from exact logical perfect-space fractions to normalized persistence.

5 Exact histories, a fixed clique palette, and hardness theorems

Let

$$G_2 = \{X, CX, CCX, H \otimes H\}.$$

Following Rudolph’s gate-dependent definition [24, Definition 2.2],¹ $\text{BQP}_1^{G_2}$ consists of promise problems decided by a polynomial-time uniform family of exact G_2 circuits on an all-zero input, with one computational-basis output bit, perfect completeness, and inverse-polynomial rejection probability in the no case. Rudolph’s exact error reduction [24, Theorem 2.3] preserves perfect completeness and stays in any gate set containing G_2 ; we apply it first so that soundness is at most $1/3$. The superscript is essential: no approximate compilation or gate-independent identification is used below.

5.1 Rational histories and the certified palette

Write $K = \sqrt{2}H$. Replace each $H \otimes H$ transition by a gain edge K on one work bit followed by a loss edge $K^{-1} = H/\sqrt{2}$ on the other. Their composite is exactly $H \otimes H$, and each edge is enforced by two orthonormal rational three-term vectors. The vectors and their direct calculation appear in Appendix B.1. This construction is due to Rudolph [24, Appendix D.1].

¹ Definition, theorem, and appendix numbers for [24] refer to the full version, arXiv:2411.02681v2.

The propagation constraints for X, CX, CCX decompose, on their constant-size active support, into two-time vectors

$$(|0, z\rangle - |1, Uz\rangle)/\sqrt{2}.$$

Clock, input, and output penalties are basis projectors. Unaffected work qubits are identities and are implemented by the harmonic padding of Lemma 3.2; they are not expanded into exponentially many rank-one terms.

► **Lemma 5.1** (Certified fixed palette). *Every local constraint in the split G_2 history is implemented by a member of one fixed family of weighted clique gadgets of total support locality at most six. Every member satisfies Assumption 3.1 and Equation (6), with uniform constants, and the graph construction is polynomial in the circuit size.*

Proof. See Appendix C. It records the finite integer certificates, exact relabelings, selected-cycle guard closure, and polynomial-size bookkeeping. ◀

5.2 Complete history kernels

Let L be the number of legal clock positions after splitting the Hadamard transitions. Put $s_t = \sqrt{2}$ immediately after a gain edge and $s_t = 1$ otherwise. If V_t is the unitary prefix after removing these scalars, then

$$\mathcal{V}v = Z^{-1/2} \sum_{t=0}^{L-1} s_t |t\rangle V_t v, \quad Z = \sum_t s_t^2, \quad L \leq Z \leq 2L. \quad (17)$$

Every gain edge is immediately followed by its matching loss edge, with no gate between them, so $s_0 = s_{L-1} = 1$.

► **Proposition 5.2** (History kernels and gaps). *Let C be the valid clean-input space, $D = \dim C > 0$, and let $M = J^* U^* P_{\text{acc}} U J$ be the compressed acceptance operator. Put $S = \ker(I - M)$. The initial history Hamiltonian has complete kernel $\mathcal{V}C$, while adding the final rejection projector gives complete kernel $\mathcal{V}S$. If $M|_{S^\perp} \preceq I/3$, then*

$$g_{\text{in}} \geq \frac{1}{8L^2}, \quad g_{\text{out}} \geq \frac{1}{3L(8L^2 + 1)} \geq \frac{1}{27L^3}. \quad (18)$$

The second statement includes $S = 0$.

The proof in Appendix B uses a weighted path Poincaré inequality. The output compression is exactly

$$\mathcal{V}^* H_{\text{rej}} \mathcal{V} = (I - M)/Z, \quad (19)$$

and a two-positive-operator estimate turns this angle penalty into the second gap in Equation (18). Combining Theorem 3.4, Theorem 4.2, Lemma 5.1, and Proposition 5.2 yields weighted clique complexes $X_{\text{in}} \subseteq X_{\text{out}}$ with

$$\beta_d(X_{\text{in}}) = D, \quad \beta_d(X_{\text{out}}) = \beta_d(X_{\text{in}} \rightarrow X_{\text{out}}) = \dim S, \quad (20)$$

and inverse-polynomial gaps above both complete kernels.

5.3 A general source: perfect-subspace fractions

Proposition 5.2 places no restriction on the clean-input space C beyond $D > 0$, and Equation (20) turns any exact circuit with a perfectly accepted subspace and a soundness bound on its complement into a normalized-persistence instance. We make this source explicit.

► **Definition 5.3** (Perfect-subspace fraction problems). *Let G be a finite gate set and let $a, b : \mathbb{N} \rightarrow [0, 1]$ satisfy $a - b \geq 1/\text{poly}$. A promise problem L is in $\text{SF}^G(a, b)$ if there are polynomials c, r and a polynomial-time uniform family of exact G -circuits U_x acting on $c(|x|)$ clean qubits, initialized to a fixed computational-basis string, and $r(|x|)$ unconstrained qubits, with a computational-basis acceptance projector P_{acc} on one output qubit, such that the compressed acceptance operator $M_x = J^* U_x^* P_{\text{acc}} U_x J$ on the unconstrained register and its perfect subspace $S_x = \ker(I - M_x)$ satisfy*

- (i) $M_x|_{S_x^\perp} \preceq \frac{1}{3}I$, that is, every state orthogonal to the perfectly accepted subspace is accepted with probability at most $1/3$;
- (ii) $\dim S_x/2^r \geq a(|x|)$ if $x \in L_{\text{yes}}$, and $\dim S_x/2^r \leq b(|x|)$ if $x \in L_{\text{no}}$.

Three sources of this type matter here.

■ *Perfect-completeness circuits.* $\text{BQP}_1^{G_2} \subseteq \text{SF}^{G_2}(3/4, 1/8)$ with $r = 3$: given a $\text{BQP}_1^{G_2}$ verifier Q_x with acceptance probability p_x , add a three-bit unconstrained label z , run Q_x unconditionally, and accept according to

$$[z = 000] \vee (q \wedge [z \in \{001, 010, 011, 100, 101\}]),$$

where q is the output of Q_x . The reversible predicate uses only X, CX, CCX , preserves z , and uncomputes its scratch. Its compressed acceptance operator on the label is exactly

$$M_x = \text{diag}(1, p_x, p_x, p_x, p_x, p_x, 0, 0), \quad (21)$$

so the perfect subspace has dimension six when $p_x = 1$ and dimension one when $p_x \leq 1/3$, and every off-perfect eigenvalue in the latter case is at most $1/3$ (Appendix B.5).

■ *One clean qubit.* Lowe et al. define SDQC_1^G [20, Definition 2] by a uniform G -circuit on one clean and $q - 1$ maximally mixed qubits, a subspace S_x of the mixed register accepted with probability exactly one, soundness $1/3$ on $S_x^\perp$, and a decision condition on the acceptance probability of the maximally mixed input. Conditions (i)–(ii) of their definition give $S_x = \ker(I - M_x)$ and $M_x|_{S_x^\perp} \preceq I/3$ (Appendix B.5). If the decision condition is imposed on the fraction $\dim S_x/2^{q-1}$, which is the quantity that their Section 3.1 identifies as the power of the class and from which the instance is decided in the proofs of their Theorems 5 and 7, the resulting class, which we write $\text{SDQC}_1^G[\text{frac}]$, is contained in $\text{SF}^G(a, b)$ with $c = 1$. With the decision condition on the acceptance probability p_x of the maximally mixed state, the identity $f \leq p_x \leq (1 + 2f)/3$ for $f = \dim S_x/2^{q-1}$ (Appendix B.5) gives $\text{SDQC}_1^G(a, b) \subseteq \text{SF}^G(\frac{3a-1}{2}, b)$ whenever $b < (3a - 1)/2$.

511 ■ *Classical counting.* A Boolean formula φ on r variables, evaluated by a reversible
 512 $\{X, CX, CCX\}$ circuit on the unconstrained register with clean scratch, has M_φ
 513 diagonal with entries $\varphi(z) \in \{0, 1\}$; so S_φ is spanned by the satisfying assignments,
 514 $\dim S_\varphi = \#\varphi$, and (i) holds with $M_\varphi|_{S^\perp} = 0$.

515 ► **Theorem 5.4** (General hardness). *For every $a - b \geq 1/\text{poly}$ there is a deterministic*
 516 *polynomial-time many-one reduction from $\text{SF}^{G_2}(a, b)$ to weighted gapped normalized*
 517 *persistence (Definition 2.1) with additive error $\varepsilon < (a - b)/2$. On input x it outputs*
 518 $X_{\text{in}} \subseteq X_{\text{out}}$ *with*

$$519 \quad \beta_d(X_{\text{in}}) = 2^r, \quad \beta_d(X_{\text{in}} \rightarrow X_{\text{out}}) = \beta_d(X_{\text{out}}) = \dim S_x, \quad q_d(X_{\text{in}}, X_{\text{out}}) = \frac{\dim S_x}{2^r},$$

520 *and explicit rational bounds $E_{\text{in}}, E_{\text{out}} \geq 1/\text{poly}$ below the positive spectra of both*
 521 *endpoint Laplacians.*

522 **Proof.** Take C to be the fixed clean string tensored with the full unconstrained
 523 register, so $D = 2^r$, and let $S = S_x$. Condition (i) is the hypothesis $M|_{S^\perp} \preceq I/3$ of
 524 Proposition 5.2, so both history kernels and gaps of Equation (18) are available. Apply
 525 Equation (20). The YES and NO values of q_d are separated by $a - b$, so additive
 526 error below $(a - b)/2$ decides. Parameter and size bookkeeping: the logical gaps in
 527 Equation (18) are explicit rationals, and the fixed palette supplies hard-coded positive
 528 rational lower bounds for the constants in Theorem 3.4. Fix $\eta_0 = 1/10$; from the
 529 polynomially bounded t_{max} , choose the first integer b' for which $\lambda = 2^{-b'}$ satisfies
 530 both inequalities in Equation (8), and for each endpoint output the largest dyadic
 531 rational E_i no greater than one half of $c_2 \eta_0^2 g_i \lambda^{2p}$. Thus $b' = O(\log t_{\text{max}})$ and the binary
 532 encodings of λ and E_i have polynomial length. If the history uses n register qubits
 533 (clock and work bits), the register has $7n$ vertices and $O(n^2)$ edges, and each gadget
 534 adds a constant number of private vertices together with $O(n)$ edges to the unaffected
 535 register factors; so the weighted graph has $O(n + t_{\text{max}})$ vertices and $O(n^2 + t_{\text{max}}n)$
 536 explicitly generated edges. Hence the map outputs the two graphs, their common
 537 weight function, d , $E_{\text{in}}, E_{\text{out}}$, and the threshold $(a + b)/2$ in deterministic polynomial
 538 time. ◀

539 ► **Corollary 5.5** (One-clean-qubit sources). *Weighted gapped normalized persistence is*
 540 $\text{SDQC}_1^{G_2}[\text{frac}]$ -*hard for all thresholds $a - b \geq 1/\text{poly}$, and $\text{SDQC}_1^{G_2}$ -hard, in the sense*
 541 *of [20, Definition 2], for all thresholds with $b < (3a - 1)/2$; for $(a, b) = (2/3, 1/3)$ the*
 542 *target error $1/24$ suffices. In these instances $\beta_d(X_{\text{in}}) = 2^{q-1}$ is the full dimension of*
 543 *the mixed register.*

544 Conjecture 1 of Lowe et al. asserts SDQC_1 -hardness of this problem for a fixed
 545 universal gate set. Corollary 5.5 proves it for G_2 in the fraction form used in their own
 546 hardness proofs, and in the trace form for every threshold pair with $b < (3a - 1)/2$.
 547 Arbitrary trace thresholds cannot be handled by any reduction whose target sees only
 548 the fraction $\dim S_x / 2^{q-1}$ (Appendix B.5).

549 ► **Theorem 5.6** (Fixed-eight hardness). *There is a deterministic polynomial-time*
 550 *many-one reduction from $\text{BQP}_1^{G_2}$ to the decision version of weighted gapped normalized*

551 persistence (Definition 2.1). Every output satisfies

$$552 \quad \beta_d(X_{\text{in}}) = 8, \quad \beta_d(X_{\text{in}} \rightarrow X_{\text{out}}) = \begin{cases} 6, & x \in L, \\ 1, & x \notin L, \end{cases}$$

553 and both endpoint degree- d Laplacians have inverse-polynomial gaps above their com-
554 plete kernels. Hence the normalized values are $3/4$ and $1/8$. Additive error $1/24$, with
555 decision threshold $7/16$, distinguishes the two cases with success probability at least
556 $2/3$.

557 **Proof.** Apply Theorem 5.4 to the eight-label source Equation (21), with $r = 3$. The
558 two ratios differ by $5/8$, and $1/24 < 5/16$. ◀

559 ► **Theorem 5.7 (Gapped Betti numbers).** Consider the problem GAPPEDBETTI: given
560 a graph G with dyadic vertex weights, a degree d , and a rational $\gamma \geq 1/\text{poly}$, with
561 the promise $\text{spec}(\Delta_{\text{Cl}(G),d}) \subseteq \{0\} \cup [\gamma, \infty)$, compute $\beta_d(\text{Cl}(G))$. GAPPEDBETTI is
562 $\#P$ -hard under polynomial-time parsimonious reductions, and it is in $\#BQP$ [3], hence
563 $\#P$ -complete under weakly parsimonious reductions. The same statements hold for
564 the persistent Betti number $\beta_d(X_1 \rightarrow X_2)$ of a filtration with gapped endpoints, and
565 with exact G_2 verifiers satisfying condition (i) of Definition 5.3 in place of Boolean
566 formulas, which gives hardness, again parsimoniously, for the perfect-completeness
567 counting class $\#BQP_1$ of Cade and Crichigno [4] restricted to G_2 .

568 **Proof.** Hardness: given φ , the classical-counting source above satisfies (i), and The-
569 orem 5.4, which uses only the two-term atoms and basis cones of the palette for
570 X, CX, CCX circuits, outputs X_{out} with $\beta_d(X_{\text{out}}) = \beta_d(X_{\text{in}} \rightarrow X_{\text{out}}) = \#\varphi$, an inverse-
571 polynomial gap above the kernel, and $\beta_d(X_{\text{in}}) = 2^r$. Containment: with the promised
572 gap, phase estimation of the explicit sparse operator $\Delta_{\text{Cl}(G),d}$ at resolution below γ is
573 a BQP verifier that accepts kernel vectors with probability exponentially close to one
574 and every vector of the orthogonal complement with probability exponentially close to
575 zero, so β_d is the dimension of its accepting subspace, a $\#BQP$ function [3, 4]; and
576 $\#BQP = \#P$ under weakly parsimonious reductions [3]. ◀

577 Crichigno and Kohler prove $\#P$ -hardness of Betti numbers of clique complexes
578 without a gap promise [6], and Cade and Crichigno prove the gapped statement for
579 general chain complexes with 6-local coboundaries [4]; Theorem 5.7 combines the two
580 features for clique complexes, and, through Corollary 7.1 below, for unweighted clique
581 complexes.

582 The exact trace-versus-perfect-space obstruction and the mixed-spectator extension
583 to individual real Hadamards are recorded in Appendix B.5. Neither is needed for
584 Theorem 5.6.

585 6 The hard instances satisfy the containment conditions

586 Lowe et al. place normalized harmonic persistence in BQP under two access condi-
587 tions [20, Lemma 7]: a polynomial-size circuit preparing the uniform mixture over

588 $\mathcal{H}_1 = \ker \Delta_{1,d}$, and the *large overlap condition*, which asks that every nonzero eigen-
 589 value of $P_{\mathcal{H}_2} P_{\mathcal{H}_1} P_{\mathcal{H}_2}$ be at least $a_{\min} \geq 1/\text{poly}$ (their Definition 3). This section shows
 590 that both conditions hold on the instances produced by Theorem 5.4, with $a_{\min}$ close
 591 to one, so that on this family the problem is simultaneously hard and solvable in
 592 quantum polynomial time. The mechanism is that the transfer estimate, applied at
 593 zero energy, forces every harmonic vector to lie close to the embedded logical kernel.

594 6.1 Harmonic spaces track the logical kernels

595 ► **Lemma 6.1** (Harmonic concentration). *Under the hypotheses and parameter choice*
 596 *Equation (8) of Theorem 3.4, every unit $x \in \mathcal{H}_A = \ker \Delta_A$ satisfies $\|(I - P_{K_A})x\| \leq \eta_1$,*
 597 *where $\eta_1 = \sqrt{C} c_1 \eta \leq \eta/2$ after fixing c_1 with $Cc_1^2 \leq 1/4$.*

598 **Proof.** Equation (7) with $\langle x, \Delta_A x \rangle = 0$ gives $\|(I - P_{K_A})x\|^2 \leq Ct_A \lambda^2 \leq Cc_1^2 \eta^2$. ◀

599 ► **Lemma 6.2** (Equal dimensions). *Let $\mathcal{H}, K$ be subspaces with $\dim \mathcal{H} = \dim K < \infty$,*
 600 *and suppose $\|(I - P_K)x\| \leq s < 1$ for every unit $x \in \mathcal{H}$. Then every singular value of*
 601 *$P_K|_{\mathcal{H}}$ is at least $\sqrt{1 - s^2}$, and $\|P_{\mathcal{H}} - P_K\| \leq s$.*

602 **Proof.** For unit $x \in \mathcal{H}$, $\|P_K x\|^2 = 1 - \|(I - P_K)x\|^2 \geq 1 - s^2$, so $P_K|_{\mathcal{H}}$ is injective
 603 and, by dimension count, maps $\mathcal{H}$ onto K . The principal angles θ_i between $\mathcal{H}$ and K
 604 therefore satisfy $\cos \theta_i \geq \sqrt{1 - s^2}$, that is, $\sin \theta_i \leq s$, and $\|P_{\mathcal{H}} - P_K\| = \sin \theta_{\max}$ for
 605 subspaces of equal dimension. ◀

606 Since $\dim \mathcal{H}_A = \dim K_A$ by Equation (9), the two lemmas give

$$607 \quad \|P_{\mathcal{H}_A} - P_{K_A}\| \leq \eta_1 \quad (A \in \mathcal{A}). \quad (22)$$

608 In words: the geometric harmonic space is an η_1 -rotation of the embedded logical
 609 kernel. This is the approximate form of the isometry requirement in Conjecture 2 of
 610 Lowe et al.; the exact rank statement is Theorem 4.2.

611 6.2 Large overlap

612 ► **Theorem 6.3** (Large overlap). *For $A \subseteq B$ in $\mathcal{A}$, the operator $P_{\mathcal{H}_B} P_{\mathcal{H}_A} P_{\mathcal{H}_B}$ has*
 613 *exactly $\dim K_B$ nonzero eigenvalues, and each of them is at least $(1 - 2\eta_1)^2$. Hence the*
 614 *pair $(P_{\mathcal{H}_B}, \mathcal{H}_A)$ satisfies the large overlap condition with $a_{\min} \geq (1 - 2\eta_1)^2 \geq 1 - 2\eta$.*

615 **Proof.** Because $K_B \subseteq K_A$, $P_{K_B} P_{K_A} = P_{K_B}$. By Equation (22),

$$616 \quad \|P_{\mathcal{H}_B} P_{\mathcal{H}_A} - P_{K_B}\| \leq \|P_{\mathcal{H}_B} - P_{K_B}\| \|P_{\mathcal{H}_A}\| + \|P_{K_B}\| \|P_{\mathcal{H}_A} - P_{K_A}\| \leq 2\eta_1.$$

617 By Weyl's inequality for singular values, every singular value of $P_{\mathcal{H}_B} P_{\mathcal{H}_A}$ is within $2\eta_1$
 618 of the corresponding singular value of P_{K_B} , which is 1 with multiplicity $\dim K_B$ and 0
 619 otherwise. The rank of $P_{\mathcal{H}_B}|_{\mathcal{H}_A}$ is the rank of the induced map $H_d(X_A) \rightarrow H_d(X_B)$
 620 (Section 2), which equals $\dim K_B$ by Theorem 4.2; so exactly $\dim K_B$ singular values
 621 are nonzero, each at least $1 - 2\eta_1$. The nonzero eigenvalues of $P_{\mathcal{H}_B} P_{\mathcal{H}_A} P_{\mathcal{H}_B}$ are the
 622 squares of the nonzero singular values of $P_{\mathcal{H}_B} P_{\mathcal{H}_A}$. ◀

623 With $\eta_0 = 1/10$ as in Theorem 5.4, $a_{\min} \geq 4/5$; choosing $\eta = 1/\text{poly}$ instead gives
 624 $a_{\min} \geq 1 - 1/\text{poly}$ at the price of a smaller, still inverse-polynomial, gap Equation (12).

6.3 Preparing the initial harmonic mixture

Let $\rho_{K_{\text{in}}}$ and $\rho_{\mathcal{H}_{\text{in}}}$ denote the uniform mixtures (normalized projectors) over $K_{\text{in}} = \mathcal{VC} \subseteq C_d(X_{\text{in}})$ and over $\mathcal{H}_{\text{in}} = \ker \Delta_{\text{in}}$.

► **Theorem 6.4** (Preparation). *The reduction of Theorem 5.4 can additionally output a polynomial-size quantum circuit that prepares $\rho_{K_{\text{in}}}$ on the simplex-label Hilbert space of X_{in} , up to $1/\text{poly}$ synthesis error. Projecting its output onto $\mathcal{H}_{\text{in}}$ by phase estimation of Δ_{in} at resolution below E_{in} succeeds with probability at least $1 - \eta_1^2$ and returns a state $\tilde{\rho}$ with*

$$\|\tilde{\rho} - \rho_{\mathcal{H}_{\text{in}}}\|_1 \leq \frac{2\eta_1^2}{1 - \eta_1^2} + \frac{1}{\text{poly}}.$$

Proof. The circuit has three stages. First, prepare the fixed clean string, the maximally mixed state on the r unconstrained qubits, and the clock state $Z^{-1/2} \sum_t s_t |t\rangle$ in the unary encoding; the amplitudes take the two values $Z^{-1/2}$ and $\sqrt{2}Z^{-1/2}$, so this is a fixed real state on $L - 1$ qubits with a polynomial-size description. Second, for $t = 1, \dots, L - 1$ apply $U_t \otimes \Pi_{\geq t}^c + I \otimes (I - \Pi_{\geq t}^c)$, where U_t is the unitary of step t (a gain or loss edge applies the unitary H ; the scalars $\sqrt{2}$ and $1/\sqrt{2}$ are already in s_t) and $\Pi_{\geq t}^c$ projects onto clock values at least t . Exactly as in [20, Lemma 8], the clock branch $|t\rangle$ accumulates $V_t = U_t \cdots U_1$, and the output is $\mathcal{V}\rho_C\mathcal{V}^*$ with ρ_C the uniform mixture over C . Third, apply to every register qubit (clock and work) the fixed isometry $\mathbb{C}^2 \rightarrow \mathbb{C}^8$ that sends the basis state $|b\rangle$ to the normalized oriented petal cycle γ_b of its bowtie, written on the labels of the eight edges of that bowtie; the tensor product of these isometries is the computational-basis embedding $(\mathbb{C}^2)^{\otimes n} \cong V \subseteq C_d(R) \subseteq C_d(X_{\text{in}})$ of Section 2, up to the orientation signs of join simplices, which are computed classically from the vertex order and applied as phases. The result is $\rho_{K_{\text{in}}}$.

For the projection, $\text{Tr}(P_{\mathcal{H}_{\text{in}}}\rho_{K_{\text{in}}}) = \frac{1}{D} \sum_{i=1}^D \cos^2 \theta_i \geq 1 - \eta_1^2$ by Lemma 6.2, where θ_i are the principal angles between $\mathcal{H}_{\text{in}}$ and K_{in} . The post-measurement state is $\sum_i \cos^2 \theta_i |h_i\rangle\langle h_i| / \sum_j \cos^2 \theta_j$ in the orthonormal eigenbasis (h_i) of $P_{\mathcal{H}}P_KP_{\mathcal{H}}$ on $\mathcal{H}_{\text{in}}$; its eigenvalues lie in $[(1 - \eta_1^2)/D, 1/((1 - \eta_1^2)D)]$, so its trace distance to $\rho_{\mathcal{H}_{\text{in}}}$ is at most $2\eta_1^2/(1 - \eta_1^2)$. The Laplacian Δ_{in} is an explicit sparse operator on the simplex labels with efficiently computable entries, and its phase estimation at inverse-polynomial resolution is the standard primitive of quantum topological data analysis [19, 11, 22]; the $1/\text{poly}$ term collects these approximation errors. ◀

6.4 Consequences

► **Theorem 6.5** (Hard and in BQP). *Weighted gapped normalized persistence remains SF^{G_2} -hard, and hence $\text{BQP}_1^{G_2}$ -hard and $\text{SDQC}_1^{G_2}[\text{frac}]$ -hard, when restricted to instances satisfying*

- (a) *the large overlap condition for $(P_{\mathcal{H}_2}, \mathcal{H}_1)$ with $a_{\min} \geq 1 - 1/\text{poly}$, and*
- (b) *the existence of a polynomial-size circuit, computable in polynomial time from the source instance, preparing a state within trace distance $1/\text{poly}$ of the uniform mixture over $\mathcal{H}_1$.*

On instances satisfying (a) and (b) the problem is in BQP by [20, Lemma 7], whose proof is linear in the prepared state and therefore tolerates the preparation error in (b).

Proof. Theorem 5.4 with $\eta = 1/\text{poly}$, Theorem 6.3, and Theorem 6.4. $\blacktriangleleft$

This is the format in which Lowe et al. state their own hardness results (“hard even when the uniform mixture can be prepared and the large overlap condition holds”, their Theorems 5, 7 and 8), and it is the standard evidence for an exponential quantum advantage in this literature [9, 10]. The unweighted version is given in Section 7.

7 A natural unweighted corollary

The weighted construction uses only weights 1 and a dyadic inverse-polynomial λ . Hayakawa’s common-copy construction replaces each vertex v by a clique of f_v labeled copies and each base edge by all cross-block edges [12]. Choose one common $F = \lambda^{-2}$ and

$$f_v = Fw(v)^2.$$

Register vertices then have F copies and private vertices have one.

In degree k , let $U_{i,k}$ map a normalized base simplex to the normalized average of its one-copy-per-block lifts. Lemma 4.2 and Theorem 4.3 of Hayakawa [12] give the symmetric identity and asymmetric gap over real chains; scalar extension gives the same statements over $\mathbb{C}$:

$$\widehat{\Delta}_{i,k} U_{i,k} = F U_{i,k} \Delta_{i,k}, \quad \widehat{\Delta}_{i,k}|_{\text{ran}(U_{i,k})^\perp} \succeq \min_v f_v I. \quad (23)$$

Use the same labeled copy block for every shared vertex at both filtration levels. With consistent orientations, simplicial inclusion then satisfies

$$\widehat{J}_{ij} U_{i,d} = U_{j,d} J_{ij}. \quad (24)$$

There is no extra normalization factor because an inclusion preserves the simplex degree, weights, labels, and copy set.

► **Corollary 7.1** (Unweighted normalized persistence). *The reduction in Theorem 5.6 maps in polynomial time to nested unweighted clique complexes with the same endpoint Betti numbers, persistent rank, and normalized values $3/4$ and $1/8$. Both unweighted endpoint Laplacians retain inverse-polynomial gaps above their complete kernels.*

Proof. The asymmetric bound in Equation (23) excludes asymmetric harmonic vectors, so

$$\ker \widehat{\Delta}_{i,d} = U_{i,d} \ker \Delta_{i,d}.$$

Together with Equation (24), this makes every inclusion-induced map on harmonic spaces unitarily equivalent to its weighted counterpart. Endpoint nullities and induced ranks are therefore preserved. If E_i lower-bounds the positive weighted spectrum, then

$$\gamma_+(\widehat{\Delta}_{i,d}) \geq \min\{F E_i, \min_v f_v\} = \min\{F E_i, 1\}.$$

By Equation (12), $FE_i = \Omega(\eta^{10} g_i / t_{\max}^8)$. Since F is polynomial, so is the blown-up graph. More explicitly, if the weighted graph has $O(n + t_{\max})$ vertices and $O(n^2 + t_{\max}n)$ edges, a safe explicit bound after blow-up is $O(F(n + t_{\max}))$ vertices and $O(F^2(n^2 + t_{\max}n))$ edges; labels and edges are generated by nested loops in polynomial time. Future private vertices may be present initially as isolated vertices; the target degree is positive, so they add no target-degree chain. ◀

The operator decomposition is Hayakawa's; the filtration-specific step is the commuting identity Equation (24). We do not count the unweighting mechanism as a separate contribution.

► **Corollary 7.2** (Unweighted general hardness and containment conditions). *Theorems 5.4, 5.7, and 6.5 and Corollary 5.5 hold verbatim for unweighted clique complexes.*

Proof. The blow-up preserves β_d at both levels and the induced rank, and scales the gap floors, exactly as in the proof of Corollary 7.1. For the containment conditions, write $\hat{\mathcal{H}}_i = U_{i,d} \mathcal{H}_i$. Since $\hat{J}U_{\text{in}} = U_{\text{out}} J$,

$$P_{\hat{\mathcal{H}}_{\text{out}}} P_{\hat{\mathcal{H}}_{\text{in}}} = U_{\text{out}} (P_{\mathcal{H}_{\text{out}}} P_{\mathcal{H}_{\text{in}}}) U_{\text{out}}^*,$$

so the large-overlap constant of Theorem 6.3 is unchanged. The preparation circuit of Theorem 6.4 is composed with the lift isometry $|\sigma\rangle \mapsto \bigotimes_{v \in \sigma} f_v^{-1/2} \sum_{i \leq f_v} |v_i\rangle$, which is exact because $F = 2^{2b'}$ is a power of two: each register vertex label receives $2b'$ copy-index qubits in the state $|+\rangle^{\otimes 2b'}$, and private vertices have one copy. ◀

8 Relation to prior work

Normalized persistence.

Lowe et al. [20] define normalized harmonic persistence (their Problem 9, our Definition 2.1), prove that a low-energy relaxation of it is DQC_1 -hard and in BQP , prove SDQC_1 -hardness of the local-Hamiltonian analogue, and conjecture SDQC_1 -hardness of Problem 9 (their Conjecture 1). Their Conjecture 2 asks for a polynomial-time map from history Hamiltonians H_1 and $H_2 = H_1 + \lambda H_{\text{out}}$ to weighted clique complexes $X_1 \subseteq X_2$ with two properties: kernel fidelity and gap, $\beta_d(X_i) = \dim \ker H_i$ together with computable inverse-polynomial gap bounds; and functoriality, $\beta_d(X_1 \rightarrow X_2) = \dim \text{ran}(P_{\ker H_2} |_{\ker H_1})$. Their Lemma 11 shows that Conjecture 2 implies Conjecture 1. Theorems 3.4 and 4.2 establish both properties for the exact G_2 history Hamiltonians of Section 5, with H_{out} at unit coefficient rather than at a small coefficient λ , with the induced map computed through the quotient model, and with the harmonic encodings shown a posteriori to be $\eta/2$ -approximate isometries compatible with the inclusion (Equation (22)). Corollary 5.5 then proves Conjecture 1 over G_2 in the fraction form. The distinction between the two forms is this: Definition 2 of Lowe et al. imposes the decision condition on the acceptance probability p_x of the maximally mixed state, whereas the quantity estimated by the oracle in the proofs of their Theorems 5 and 7 is the fraction $f_x = \dim S_x / 2^{q-1}$, and after the Marriott–Watrous amplification used there the acceptance probability of the amplified circuit is within $1/\text{poly}$ of f_x (their

equation (9)) but no longer within $1/\text{poly}$ of p_x . Since $f_x \leq p_x \leq (1 + 2f_x)/3$ (Appendix B.5), the trace promise implies the fraction promise exactly when $b < (3a - 1)/2$, and Corollary 5.5 covers those thresholds in the trace form as well; for other thresholds no reduction whose target depends only on S_x can exist. Their containment result (Lemma 7) needs a preparation circuit for the uniform mixture over $\ker \Delta_{1,d}$ and an inverse-polynomial lower bound on the nonzero eigenvalues of $P_{\mathcal{H}_2} P_{\mathcal{H}_1} P_{\mathcal{H}_2}$; Section 6 proves both on our instances, which is why Theorem 6.5 can be stated in the same form as their Theorems 5, 7 and 8.

Clique homology gadgets.

Crichigno and Kohler [6] give a parsimonious reduction in which satisfying states correspond one-to-one to holes, so exact multiplicity by itself is not new; their instances carry no gap promise. King and Kohler [16] introduce the weighted gadgets and the many-gadget spectral analysis behind gapped clique homology, proving a gap when the logical Hamiltonian is positive definite, with a gadget weight proportional to the logical gap; Hayakawa [12] removes the weights under the same positive-definite promise. Our transfer theorem is formulated from finite zero-weight certificates, applies to arbitrary geometric chains and to degenerate kernels, decouples λ from g , and takes its exponent from the fillings rather than from the locality. Its padding step differs from the coordinate-bulk formulation displayed in the arXiv version of King and Kohler [15, Claim 10.2]: there the bulk consists of all coordinates containing a private vertex, which in the padded complex also contains boundary terms obtained by deleting a weight-one outside vertex, whereas we keep local private coordinates tensored with outside harmonics in every bidegree (Lemma 3.2). We also work with the fixed local matrices before tensoring with the outside identity, so that no operator bound on a projector of growing rank has to be inferred from basiswise perturbation estimates. These are differences of proof technique, and we make no claim about the presentation in the published version [16].

Counting Betti numbers.

Crichigno and Kohler prove that computing Betti numbers of clique complexes is $\#P$ -hard [6, Theorem 4], through the $\#P$ -hardness of counting quantum 4-SAT solutions; their instances have no gap promise. Schmidhuber and Lloyd prove $\#P$ -hardness of exact Betti numbers and NP -hardness of multiplicative approximation for clique complexes, even in the clique-dense regime where the quantum algorithms perform best [25]; the restriction is a density condition, not a spectral one. Cade and Crichigno prove $\#P$ -completeness with an inverse-polynomial gap for general chain complexes with 6-local coboundaries [4, Theorem 5], using the counting class $\#BQP$ of Brown, Flammia, and Schuch [3], whose definition includes a gap promise on the acceptance operator and which equals $\#P$ under weakly parsimonious reductions. Theorem 5.7 is the clique-complex statement with the gap, for weighted and unweighted graphs; it is a direct corollary of the degenerate-kernel transfer, since the classical source has $M_\varphi|_{S^\perp} = 0$.

Persistence hardness.

Gyurik et al. [10] prove BQP_1 -hardness, over the gate set $\{\text{CNOT}, U_{\text{Pyth}}\}$, of deciding whether a *given* harmonic representative of X_1 survives in X_2 , with a gap promise on the final Laplacian. Normalized persistence asks instead for the fraction of the full initial homology that survives, with no representative given, so a hardness proof must control the complete kernels at both endpoints. Persistent Laplacians [23] give a spectral formulation of persistent Betti numbers; we use ordinary endpoint Laplacians for the promise and the quotient model for the map.

Sources and gate sets.

Rudolph [24] defines the gate-dependent classes, proves exact error reduction inside gate sets containing G_2 , shows $\text{BQP}_1^G \subseteq \text{BQP}_1^{G_2}$ for every finite G with entries in some $\mathbb{Q}(\zeta_{2^k})$, and supplies the gain/loss split of $H \otimes H$ with rational three-term constraint vectors and the base clique gadgets used in our palette. Cade and Crichigno [4] relate cohomology problems to QMA_1 and DQC_1 , and Gyurik, Cade, and Dunjko [9] connected quantum topological data analysis to DQC_1 [18]. The history-state construction with a unary clock and neighbor guards is Kitaev's [17, 14], with the rational amplitude bookkeeping of Section 5; the amplification used by Lowe et al. is that of Marriott and Watrous [21].

Unweighting.

Hayakawa's symmetric/asymmetric decomposition of the common-copy blow-up [12, Lemma 4.2 and Theorem 4.3] is used as a black box. Our only addition is the observation that identical labeled copy blocks at both filtration levels make the symmetric isometries commute with inclusion, Equation (24), which also transfers the containment conditions. We do not count this as a separate contribution.

9 Limitations and open problems

The construction has three substantive limits. Its exact palette is finite and real: arbitrary integer states and complex phases are outside the theorem, so every hardness statement is relative to the gate set G_2 (extended to cyclotomic gate sets only for BQP_1 , through Rudolph's simulation). Trace-form SDQC_1 -hardness is proved only for threshold pairs with $b < (3a - 1)/2$, and Appendix B.5 shows that this is the limit of any reduction whose target sees only the perfectly accepted subspace. Finally, the gap is a complexity promise rather than a practical estimate: at depth five it scales conservatively as $\Omega(\eta^{12}g/t^{10})$ with $g = \Omega(L^{-3})$, and the common-copy blow-up multiplies the vertex count by $F = \lambda^{-2}$.

Each limit suggests a question.

1. Does SF^G -hardness extend to every finite gate set $G \subseteq \mathbb{Q}(\zeta_{2^k})$? Rudolph's simulation accepts when its postselection fails; on an SF verifier this keeps S_x perfectly accepted and, if the failure probability is input-independent and exponentially

- 817 small, keeps the soundness on $S_x^\perp$ at $1/3 + 2^{-\text{poly}}$. Verifying these two properties
 818 would remove most of the gate dependence.
- 819 **2.** Is there an exact transformation of an SDQC_1 verifier whose perfect fraction encodes
 820 the acceptance probability of the maximally mixed state? If not, the fraction form
 821 is the right definition of the class, and Conjecture 1 of Lowe et al. should be read
 822 in that form.
 - 823 **3.** Can the depth be reduced? A palette whose fillings use at most two private vertices
 824 per simplex would give $\kappa = 4$ and $E = \Omega(\eta^6 g/t^4)$.
 - 825 **4.** Lowe et al. ask for a general kernel-preserving universality theorem for weighted
 826 clique Laplacians. Theorem 3.4 is such a theorem for palettes with finite certificates.
 827 Extending the certificate model to integer-state gadgets [16] or to complex phases
 828 would answer their question for a much wider class of Hamiltonians.
 - 829 **5.** Do the hard instances also have inverse-polynomial gaps of the persistent Lapla-
 830 cian [23]? This is not needed for Theorem 6.5, but it would connect the instances
 831 to the persistent Betti number algorithms of [11, 22].

832 Within its limits, the paper establishes a reusable criterion: a new clique reduction
 833 obtains exact kernel multiplicity, an inverse-polynomial gap above the whole kernel,
 834 harmonic spaces that track the logical kernels, and natural persistent ranks by certifying,
 835 for a finite palette, local zero-weight kernels, private coercivity, exact filling, and
 836 interfaces, with no growing-dimensional perturbation argument.

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918 **A Proof of the all-chain transfer theorem**

919 This appendix proves Theorem 3.4 from the finite local hypotheses. All constants
920 below depend only on the fixed palette.

921 **A.1 Harmonic padding**

922 For padding, use augmented reduced chain complexes, including degree -1 , in every
923 join bidegree:

$$924 \quad \tilde{C}_d(Y_a * R_{\text{out}}) = \bigoplus_{r+s=d-1} \tilde{C}_r(Y_a) \otimes \tilde{C}_s(R_{\text{out}}).$$

925 On each summand the join Laplacian is the tensor sum

$$926 \quad \Delta_{Y_a * R_{\text{out}}} = \Delta_{Y_a} \otimes I + I \otimes \Delta_{R_{\text{out}}}.$$

927 The outside register has reduced homology only in its top degree and a constant
928 positive gap on every other mode. At the global target degree, therefore, the only
929 outside-harmonic bidegree is the active target degree tensored with $\mathcal{H}_{\text{out}}$. This proves
930 the stated zero-weight kernel and puts every other bidegree above a uniform floor.
931 Tensor the private output projector with $P_{\mathcal{H}_{\text{out}}}$; both outside differentials vanish after
932 this projection, preserving the private-pair bounds.

933 Finally, the relative join formula identifies the padded connecting map with the
934 active connecting map tensored with $\mathcal{H}_{\text{out}}$. Since all register weights are one, the
935 positive-weight diagonal gauge fixes the register coordinates. Thus the padded bound-
936 ary intersection is $\text{ran}(\pi_a \otimes I_{\mathcal{H}_{\text{out}}})$, completing the proof of Lemma 3.2.

937 **A.2 Global decomposition and interface energy**

938 Write the target chain space as the orthogonal coordinate sum

$$939 \quad C_d(X_A) = C_d(R) \oplus \bigoplus_{j \in A} C_{d,j}^{\text{priv}}.$$

940 For a unit chain x , write

$$941 \quad x = \omega_0 + \sum_j \omega_j, \quad x_j = \omega_0 + \omega_j, \quad U = \sum_j \|\omega_j\|^2, \quad e = \langle x, \Delta_A x \rangle.$$

942 Let $a = \partial_R \omega_0$, $y_j = L_j \omega_j$, $y = \sum_j y_j$, and $r = a + y$. The vector r is the shared-register
943 component of the global down-boundary. Since $a + y_j = r - (y - y_j)$,

$$944 \quad \sum_j \|a + y_j\|^2 \leq 2t_A \|r\|^2 + 2 \sum_j \|y - y_j\|^2.$$

945 The elementary identity

$$946 \quad \sum_j \|y - y_j\|^2 = (t_A - 2) \|y\|^2 + \sum_j \|y_j\|^2$$

also holds for $t_A = 1$. Put

$$\rho = \lambda^{-2} \| [L_1 \cdots L_{t_A}] \|^2 \leq N_* t_A, \quad \chi = (t_A - 2)_+ \rho + N_* \leq C t_A^2.$$

Then Equation (6) gives

$$\sum_j \|y - y_j\|^2 \leq \chi \lambda^2 U.$$

All nonregister down-boundary coordinates and all up-boundary coordinates are private and partition by gadget. We obtain

$$\sum_j \|D_j(\lambda)x_j\|^2 \leq 2t_A e + 2\chi \lambda^2 U. \quad (25)$$

A.3 Zero-weight and private-sector coercivity

Let Q_j also denote the orthogonal projector onto the padded zero-weight private subspace $Q_j \otimes \mathcal{H}_{\text{out}}$ of gadget j , and let $A_{j,0} = I - P_V - Q_j$ in the padded local chain space; by Lemma 3.2 this is the projector onto the orthogonal complement of $\ker D_{j,0}$. The positive singular-value floor of $D_{j,0}$ on $\text{ran } A_{j,0}$, together with $D_j(\lambda) = D_{j,0} + \lambda D_{j,1}$, gives

$$\begin{aligned} S_A &:= \sum_j \|A_{j,0}x_j\|^2 \leq C \sum_j \|D_{j,0}x_j\|^2 \\ &\leq C \left(t_A e + \chi \lambda^2 U + \lambda^2 \sum_j \|x_j\|^2 \right) \\ &\leq C \left(t_A e + \chi \lambda^2 U + t_A \lambda^2 \right). \end{aligned} \quad (26)$$

The final inequality uses $\sum_j \|x_j\|^2 = t_A \|\omega_0\|^2 + U \leq t_A$.

Because $T_j P_V = 0$,

$$T_j Q_j x_j = T_j x_j - T_j A_{j,0} x_j.$$

The lower bound on T_j restricted to Q_j , the estimate $\|T_j\| \leq C\lambda$, and orthogonality of the retained private output blocks imply

$$S_Q := \sum_j \|Q_j x_j\|^2 \leq C \left(\frac{e}{\lambda^2} + S_A \right). \quad (27)$$

The orthogonality here concerns the retained T_j outputs. The shared L_j outputs were handled by their horizontal operator norm and need not be orthogonal.

Let $N = \|(I - P_V)x\|^2$. Summing the orthogonal decompositions $P_V \oplus Q_j \oplus \text{ran } A_{j,0}$ gives

$$N \leq U + t_A \|(I - P_V)\omega_0\|^2 = S_A + S_Q.$$

Since $U \leq N$, substituting Equations (26) and (27) and using $\chi \leq C t_A^2$ yields

$$N \leq C \left[t_A \lambda^2 + (t_A + \lambda^{-2})e \right] + C t_A^2 \lambda^2 N.$$

Choosing $\lambda \leq c_0/t_{\max}$ absorbs the last term and proves Equation (10).

A.4 Logical coercivity from the fillings

► **Lemma A.1** (Schur-complement filling bound). *Let $A \succeq 0$, let $x \in \text{ran } A$, and let u satisfy $\langle x, A^+x \rangle \leq \|u\|^2$, where A^+ is the pseudo-inverse. Then $\langle y, Ay \rangle \geq |\langle x, y \rangle|^2 / \|u\|^2$ for every y . For $A = \partial\partial^*$ and $x = \partial u$, the hypothesis holds, since $\langle x, (\partial\partial^*)^+x \rangle = \min\{\|u'\|^2 : \partial u' = x\}$.*

Proof. Cauchy–Schwarz: $|\langle x, y \rangle|^2 = |\langle A^{+1/2}x, A^{1/2}y \rangle|^2 \leq \langle x, A^+x \rangle \langle y, Ay \rangle$. For the second statement, $(\partial\partial^*)^+ = (\partial^*)^+\partial^+$ on $\text{ran } \partial$, so $\langle x, (\partial\partial^*)^+x \rangle = \|\partial^+x\|^2$, and ∂^+x is the minimum-norm preimage. ◀

Fix a padded gadget j of palette type a , with integer register cycle v_a , integer filling c_a , and depth p_a (Definition 3.3). In weighted coordinates the filling of v_a is $W_{d_a+1}^{-1}c_a$, because W is the identity on register coordinates, and

$$\|W_{d_a+1}^{-1}c_a\|^2 = \sum_{\sigma} c_a(\sigma)^2 \lambda^{-2\text{priv}(\sigma)} \leq \|c_a\|^2 \lambda^{-2p_a} \quad (0 < \lambda \leq 1).$$

Every unit vector of $\text{ran } \Pi_j$ has the form $x = (v_a/\|v_a\|) \otimes h$ with h a unit vector of $\mathcal{H}_{\text{out}}$, and $u = (W_{d_a+1}^{-1}c_a/\|v_a\|) \otimes h$ satisfies $\partial u = x$ in the padded complex, because outside weights are one and $\partial h = 0$; its norm is $\|W^{-1}c_a\|/\|v_a\|$. Applying Lemma A.1 to $A = \Delta_j^\uparrow$ and $x = \Pi_j y / \|\Pi_j y\|$ gives, for every y ,

$$\langle y, \Delta_j^\uparrow y \rangle \geq \frac{\|\Pi_j y\|^2 \|v_a\|^2}{\|W^{-1}c_a\|^2} \geq \frac{\|v_a\|^2}{\|c_a\|^2} \lambda^{2p_a} \|\Pi_j y\|^2,$$

that is, $\Delta_j^\uparrow \succeq c\lambda^{2p}\Pi_j$ with the palette constant $c = \min_a \|v_a\|^2/\|c_a\|^2$. The guard construction of Appendix C fills $v \otimes \gamma_\beta$ by $c_a \otimes \gamma_\beta$, and padding tensors with register cycles, so neither changes the depth. The top-dimensional register has no up-columns, and every degree- $(d+1)$ simplex with a private vertex belongs to exactly one gadget (Definition 2.2). Hence the global up-Laplacian is the exact sum of the embedded local up-Laplacians, proving Equation (11).

Since $K_A \subseteq V$, the squared distance decomposes as

$$\|(I - P_{K_A})x\|^2 = N + \langle x, (P_V - P_{K_A})x \rangle.$$

Equations (5) and (11) give

$$\langle x, (P_V - P_{K_A})x \rangle \leq \frac{e}{cg_A\lambda^\kappa}.$$

Together with Equation (10),

$$\|(I - P_{K_A})x\|^2 \leq C \left[t_A \lambda^2 + (t_A + \lambda^{-2})e + \frac{e}{g_A \lambda^\kappa} \right].$$

For $\lambda \leq c_0/t_{\max}$, $g_A \leq 1$, and $\kappa = 2p \geq 2$, the last energy term dominates the middle energy terms up to a palette constant. This proves Equation (7).

A.5 Exact multiplicity and the whole-positive-spectrum gap

Let $\mathcal{L}_A(E_A)$ be the span of all eigenvectors of Δ_A with eigenvalue below E_A . Every unit $x \in \mathcal{L}_A(E_A)$ obeys $e < E_A$. Choosing the constants in Equation (8) makes the right-hand side of Equation (7) strictly less than $\eta^2 < 1$. Therefore $P_{K_A}x \neq 0$ for every nonzero $x \in \mathcal{L}_A(E_A)$, so $P_{K_A}|_{\mathcal{L}_A(E_A)}$ is injective and

$$\dim \mathcal{L}_A(E_A) \leq \dim K_A. \quad (28)$$

Every $v \in V$ remains a cycle after attaching the gadgets. By Lemma 4.1, the map $V/W_A \rightarrow H_d(X_A)$ is injective. Its domain has dimension $\dim K_A$, so Hodge theory supplies at least $\dim K_A$ zero eigenvectors of Δ_A . These zero vectors belong to $\mathcal{L}_A(E_A)$. Combining this lower bound with Equation (28) forces equality throughout: the geometric kernel has dimension $\dim K_A$, and no positive eigenvalue lies below E_A .

If $K_A = 0$, any unit vector in $\mathcal{L}_A(E_A)$ would have distance one from K_A , contradicting the strict $\eta < 1$ bound. Thus the same argument proves $\Delta_A \succeq E_A I$ in the zero-kernel case. No additional global relative-acyclicity assumption is required.

A.6 Proof of quotient naturality

Equation (13) makes q_A well-defined and injective. Its domain has dimension

$$\dim(V/W_A) = \dim K_A.$$

Hodge theory and Equation (9) give

$$\dim H_d(X_A) = \dim \ker \Delta_A = \dim K_A,$$

so q_A is also surjective. If $A \subseteq B$, both paths in the diagram of Theorem 4.2 send $v + W_A$ to the class of the same register cycle v in X_B . Thus the diagram commutes. The quotient map is onto because $W_A \subseteq W_B$, and its rank is $\dim(V/W_B) = \dim K_B$.

B History-gap and source calculations

B.1 Rational gain/loss split

On a clock bit c and work bit w , define

$$\begin{aligned} u_0 &= (|00\rangle + |01\rangle - |10\rangle)/\sqrt{3}, & u_1 &= (|00\rangle - |01\rangle - |11\rangle)/\sqrt{3}, \\ d_0 &= (|00\rangle - |10\rangle - |11\rangle)/\sqrt{3}, & d_1 &= (|01\rangle - |10\rangle + |11\rangle)/\sqrt{3}. \end{aligned}$$

The vectors in each pair are orthonormal. If consecutive history amplitudes are f_t, f_{t+1} , orthogonality to u_0, u_1 is equivalent to $f_{t+1} = \sqrt{2}Hf_t$; orthogonality to d_0, d_1 is equivalent to $f_{t+1} = Hf_t/\sqrt{2}$. Applying the first relation to one work bit and the second to the other gives $H \otimes H$.

1039 B.2 Weighted path estimate

1040 With the unary encoding $|t\rangle = |1^t 0^{T-t}\rangle$, the legal clock span and its orthogonal
 1041 complement are invariant under the neighboring-clock-guarded propagation, input,
 1042 and output projectors. The diagonal clock term

$$1043 \quad H_{\text{clock}} = \sum_{j=1}^{T-1} |01\rangle\langle 01|_{j,j+1}$$

1044 has energy at least one on every illegal computational-basis clock string. By Hermiticity,
 1045 the whole unary-clock Hilbert space is the orthogonal sum of the invariant legal and
 1046 illegal sectors. It therefore suffices to prove the smaller inverse-polynomial bound
 1047 on the legal sector. For a legal-clock state, write $f_t = s_t V_t \xi_t$, with the notation
 1048 of Equation (17). Its norm and initial-history energy are

$$1049 \quad N(\xi) = \sum_t w_t \|\xi_t\|^2, \quad E(\xi) = \sum_{t=1}^{L-1} c_t \|\xi_t - \xi_{t-1}\|^2 + \langle \xi_0, A_{\text{in}} \xi_0 \rangle,$$

1050 where $w_t = s_t^2 \in [1, 2]$, $c_t \in \{1/2, 2/3\}$, and A_{in} is the projector onto $C^\perp$. Split every
 1051 ξ_t into the clean subspace $C = \ker A_{\text{in}}$ and its orthogonal complement.

1052 For a state orthogonal to all valid histories, the clean component has weighted
 1053 mean zero:

$$1054 \quad \sum_t w_t \xi_t^C = 0.$$

1055 The weighted variance identity and path Cauchy-Schwarz inequality give

$$1056 \quad \sum_t w_t \|\xi_t^C\|^2 = \frac{1}{2 \sum_t w_t} \sum_{i,j} w_i w_j \|\xi_i^C - \xi_j^C\|^2 \leq 2L^2 E_C.$$

1057 For the dirty component, set $a = \|\xi_0^{C^\perp}\|^2$ and $D_B = \sum_t \|\xi_t^{C^\perp} - \xi_{t-1}^{C^\perp}\|^2$. The bound

$$1058 \quad \|\xi_t^{C^\perp}\|^2 \leq 2a + 2(L-1)D_B$$

1059 implies

$$1060 \quad \sum_t w_t \|\xi_t^{C^\perp}\|^2 \leq 8L^2 E_B.$$

1061 Adding both sectors proves $N(\xi) \leq 8L^2 E(\xi)$ on the orthogonal complement of the
 1062 complete initial kernel, establishing the first bound in Equation (18).

1063 B.3 Adding the output projector

1064 We record the two-positive-operator estimate used for the second bound in Equa-
 1065 tion (18). Let P project onto $\ker A$, suppose $A \succeq g(I - P)$, and let R be an orthogonal
 1066 projector. Put $\mathcal{S} = \ker A \cap \ker R$, and suppose $PRP \succeq \alpha P$ on $\ker A \ominus \mathcal{S}$. On every
 1067 nontrivial canonical block for the pair P, R , for some $\rho \in [\alpha, 1]$,

$$1068 \quad P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} \rho & \sqrt{\rho(1-\rho)} \\ \sqrt{\rho(1-\rho)} & 1-\rho \end{pmatrix}.$$

Thus $g(I - P) + R$ has trace $g + 1$, determinant $g\rho$, and smallest eigenvalue

$$\frac{g + 1 - \sqrt{(g + 1)^2 - 4g\rho}}{2}.$$

This quantity increases with ρ ; the one-dimensional canonical blocks outside $\mathcal{S}$ obey the same lower bound. Since $A + R \succeq g(I - P) + R$ and both operators have kernel $\mathcal{S}$,

$$\gamma_+(A + R) \geq \frac{g + 1 - \sqrt{(g + 1)^2 - 4g\alpha}}{2} \geq \frac{g\alpha}{g + 1}. \quad (29)$$

For the history Hamiltonian, $g = 1/(8L^2)$ and $\alpha = (1 - r)/Z \geq 1/(3L)$ when $r \leq 1/3$. Substitution into Equation (29) gives

$$\gamma_+(H_{\text{out}}) \geq \frac{1}{3L(8L^2 + 1)}.$$

The kernel identity follows directly from positivity:

$$\ker(H_{\text{in}} + H_{\text{rej}}) = \ker H_{\text{in}} \cap \ker H_{\text{rej}} = \mathcal{V} \ker(I - M).$$

B.4 Exact eight-label compression

Let U_x denote the fixed-input verifier and q its output bit. The label z is never changed. After computing and uncomputing the constant-size acceptance predicate, compression through all clean work registers has no matrix element between distinct labels. On $z = 000$, acceptance is the identity. On five labels it is the scalar $p_x = \|P_{\text{acc}} U_x |0 \cdots 0\rangle\|^2$, and on two labels it is zero. This proves Equation (21) even for a coherent vector across label values.

The normalized trace is

$$\frac{\text{Tr } M_x}{8} = \frac{1 + 5p_x}{8}.$$

If $p_x = 1$, this equals $3/4$ and coincides with the perfect-space fraction. If $p_x \leq 1/3$, it is at most $1/3$, the perfect-space fraction is exactly $1/8$, and every off-perfect eigenvalue is at most $1/3$. Thus the construction simultaneously supplies the trace promise, the perfect-space separation, and the off-perfect ceiling needed by the output-gap proof.

B.5 Source-interface boundary

The exact source used by Theorem 5.6 begins with a fixed-input $\text{BQP}_1^{G_2}$ verifier. The three mixed label bits are added by the reduction and are the entire input space of Equation (21); the verifier's input and all predicate scratch remain clean. Consequently the initial history dimension is exactly eight. No trace-to-rank inference, unrestricted SDQC_1 identification, or approximate gate compilation enters the proof.

The finite gadget evidence needed to compile these circuit terms is given in Appendix C. The theorem uses those exact finite objects and does not infer their topology from floating-point spectra.

1102 ► **Lemma B.1** (Perfect subspaces). *Let $0 \preceq M \preceq I$ on $\mathbb{C}^D$ and let S be a subspace*
 1103 *such that $\langle \psi, M\psi \rangle = 1$ for every unit $\psi \in S$ and $\langle \psi, M\psi \rangle \leq r < 1$ for every unit*
 1104 *$\psi \in S^\perp$. Then $S = \ker(I - M)$ and $M|_{S^\perp} \preceq rI$.*

1105 **Proof.** For unit $\psi \in S$, $\langle \psi, (I - M)\psi \rangle = 0$ with $I - M \succeq 0$ forces $M\psi = \psi$; so
 1106 $S \subseteq \ker(I - M)$ and S is M -invariant, hence so is $S^\perp$. If $\psi = s + s' \in \ker(I - M)$
 1107 with $s \in S$, $s' \in S^\perp$, then $Ms = s$ gives $Ms' = s'$, so $\|s'\|^2 = \langle s', Ms' \rangle \leq r\|s'\|^2$ forces
 1108 $s' = 0$. Finally $M|_{S^\perp}$ is the compression of M to an invariant subspace on which the
 1109 quadratic form is at most r . ◀

1110 For an arbitrary acceptance operator with $M|_{S^\perp} \preceq rI$, put $f = \dim S/D$ and
 1111 $p = \text{Tr}(M)/D$. Spectral decomposition gives

$$1112 \quad f \leq p \leq r + (1 - r)f, \quad \text{so for } r = \frac{1}{3}: \quad f \leq p \leq \frac{1+2f}{3}.$$

1113 Thus trace promises $p \geq a$ versus $p \leq b$ force a perfect-space fraction gap only when
 1114 $\max\{0, (a - r)/(1 - r)\} > b$, that is, $b < (3a - 1)/2$ for $r = 1/3$. The obstruction
 1115 is genuine: for any q in the overlap of the two intervals, the operators $I_{qD} \oplus 0$ and
 1116 $I_{qD} \oplus \frac{a-q}{1-q}I$ have the same perfect fraction q and satisfy opposite trace promises, and
 1117 no transformation whose output depends only on the perfect subspace can separate
 1118 them. The eight-label source avoids this obstruction by fixing the perfect fractions
 1119 directly, and Definition 5.3 imposes the promise on the fraction.

1120 An exact circuit over $\{X, CX, CCX, H\}$ admits the following real-gate extension.
 1121 Add one maximally mixed, unmeasured spectator a and replace each H_j by $H_j \otimes H_a$.
 1122 If there are h Hadamards, then

$$1123 \quad U' = U \otimes H_a^h, \quad J' = J \otimes I_a, \quad P'_{\text{acc}} = P_{\text{acc}} \otimes I_a.$$

1124 For either parity of h ,

$$1125 \quad (J')^*(U')^*P'_{\text{acc}}U'J' = M \otimes I_a.$$

1126 Both the input dimension and perfect eigenspace dimension double, while the fraction
 1127 and off-perfect ceiling are unchanged. This calculation does not cover complex phases.

1128 **C** Finite palette and guard certificates

1129 This appendix identifies the finite objects used in Lemma 5.1. The underlying active
 1130 graphs are four explicit constructors in the source artifact: `state_0m1`, `state_00`
 1131 `m11`, `state_01m10`, and `state_00m10m11`. They accompany Rudolph's gain/loss
 1132 construction [24]. The supplementary certificate archive records the complete graphs,
 1133 oriented integer fillings, kernel bases, modular rank witnesses, and Gram determinants.
 1134 No numerical eigenvalue threshold is used.

1135 For the one-qubit atom, exact boundary ranks are 15, 24 and the Betti numbers
 1136 are (1, 1, 0); a denominator-one 24-term two-chain fills the target cycle. At zero
 1137 weight the target differential pair has rank 30, and the two register cycles plus eight
 1138 private vectors give its complete ten-dimensional kernel. For $|00\rangle - |11\rangle$, the clique

Active vector	degree	vertices	dim V	dim Q	depth p_a	certificate mod p
$ 0\rangle - 1\rangle$	1	16	2	8	3	$\det G = 512$
$ 00\rangle - 11\rangle$	3	33	4	112	5	$\det G = 655970$
$ 01\rangle - 10\rangle$	3	33	4	112	5	signed relabeling
$ 00\rangle - 10\rangle - 11\rangle$	3	41	4	176	5	$\det G = 797443$
three remaining gain/loss vectors	3	41	4	176	5	signed relabelings

■ **Table 1** Exact active-palette data. Vertex counts include the $7m_a$ register vertices of the active bowties. The depth column is the largest number of private vertices in a simplex of the certified filling (Definition 3.3); the basis cone has depth 1. In each determinant row, $p = 1000003$ and $G = T_{a,1}^* T_{a,1}$ on Q_a ; a nonzero residue proves injectivity over $\mathbb{Q}$.

counts in degrees zero through four are $(33, 241, 646, 712, 272)$, the boundary ranks are $(32, 209, 437, 272)$, and the Betti numbers are $(1, 0, 0, 3, 0)$. A denominator-one 248-term four-chain fills the target; rank 596 of the zero-weight pair matches the four register and 112 private kernel vectors. Swapping the second bowtie petals transports this certificate exactly to $|01\rangle - |10\rangle$.

For the three-term representative, the clique counts in degrees zero through five are $(41, 322, 914, 1082, 477, 30)$,

and the boundary ranks are $(40, 282, 632, 447, 30)$. Hence the target degree-three Betti number is three. A denominator-one 382-term four-chain fills $|00\rangle - |10\rangle - |11\rangle$, while three complementary register cycles remain independent modulo boundaries. The zero-weight pair has rank 902, matching its kernel $V \oplus Q$ of dimension $4 + 176$. Signed permutations of bowtie petals transport the graph, filling, kernel, and private pair to the other three gain/loss vectors.

C.1 Selected-cycle guards

We record the guard closure because it is part of the theorem interface, not an appeal to a general integer-state gadget. Let Y implement π_ϕ on register R , let B be a fresh bowtie with petals S_0, S_1 , and choose $\beta \in \{0, 1\}$. Add all edges of B , join every vertex of R to every vertex of B , and join each private vertex of Y only to the selected petal S_β . The resulting flag complex is

$$Z = (Y * S_\beta) \cup (R * B), \quad (Y * S_\beta) \cap (R * B) = R * S_\beta. \quad (30)$$

Its new register is $R' = R * B$. If γ_β is the normalized oriented cycle of the selected petal, then Z implements

$$\pi_\phi \otimes |\beta\rangle\langle\beta|, \quad Q' = Q \otimes \mathbb{C}\gamma_\beta.$$

Indeed, reduced relative chains give

$$\tilde{C}_*(Z, R') \cong \tilde{C}_*(Y, R) \hat{\otimes} \tilde{C}_*(S_\beta).$$

The connecting map therefore tensors the old exact boundary intersection with $[\gamma_\beta]$, proving exact filling and no additional killed register class. At zero private weight the private differential is the corresponding tensor-sum differential. Since S_β has reduced homology only in degree one, the complete target kernel is $V' \oplus Q'$; every other bidegree pays a fixed positive guard gap. Finally, project the old private output pair onto the selected harmonic line. Both outside differentials vanish under this projection, so

$$T_Z(\lambda) = \lambda T_{Z,1}, \quad T_{Z,1}V' = 0, \quad \|T_{Z,1}(q \otimes \gamma_\beta)\| \geq b\|q \otimes \gamma_\beta\|.$$

Removing the unique old private vertex also gives $LL^* \preceq N_{\text{priv}}\lambda^2 I$. Iterating this construction for the constant number of required basis guards preserves a fixed finite palette.

The basis projector itself is the special case $R \cup (v * S_\beta)$, where a private apex v cones the selected petal. Its private zero-weight kernel Q is zero, so the private-pair condition is vacuous. Together with Table 1, this proves Lemma 5.1.

C.2 Reproducibility boundary

The supplementary archive contains the exact witnesses in

- RUDOLPH_REPRESENTATIVE_BULK_CERTIFICATE.json,
- REMAINING_ACTIVE_ATOM_CERTIFICATES.json,
- ACTIVE_HADAMARD_ORBIT_CERTIFICATE.json, and
- SELECTED_CYCLE_GUARD_CHECKS.json.

Their checkers recompute the stated ranks, fillings, relabelings, and guard identities. These finite checks certify the displayed palette; the analytic implication from those certificates is the proof of Theorem 3.4.

AI Disclosure

We used ChatGPT and Claude as auxiliary tools for literature search, brainstorming, proof checking, and drafting and editing of the exposition. The authors evaluated and revised all AI-assisted material and take full responsibility for the correctness and originality of the submission, including every proof and reference.